

Air conditioner

Installation manual

AR**TSEC** / AR**TVFC** / AR**TSHZ** / AR**TVHZ** /
AR**TSFC** / AR**TSHZ**

- Thank you for purchasing this Samsung air conditioner.
- Before operating this unit, please read this installation manual carefully and retain it for future reference.



DB68-08814A-00

SAMSUNG

Contents

Safety Information	3
Safety Information	3
Installation	4
Preparation	4
Step 1.1 Viewing the typical installation	4
Step 1.2 Choosing the installation location	5
Step 1.3 Packing and Unpacking	8
Step 1.4 Preparing materials and tools	9
Indoor Unit Installation	11
Step 2.1 Attaching the mounting bracket to the wall	11
Step 2.2 Drilling the wall penetration	11
Step 2.3 Connecting the refrigerant pipes	12
Step 2.4 Connecting the power and communication cables	13
Step 2.5 Optional: Extending the power cable	15
Step 2.6 Connecting the drain hose	16
Step 2.7 Taping the pipes, cables, and drain hose	17
Outdoor Unit Installation	18
Step 3.1 Mounting the outdoor unit	18
Step 3.2 Connecting the cables and the pipes	19
Installation Inspection and Testing	21
Step 4.1 Performing a drain leak test	21
Step 4.2 Performing the gas leak tests	21
Step 4.3 Evacuating the system	22
Step 4.4 Adding refrigerant (if needed)	23
Step 4.5 Preparing the system for commissioning	23
Step 4.6 Commissioning the unit	23
Step 4.7 Performing final checks and trial operation	24

Safety Information



WARNING: Read This Manual

- Read and follow all safety information and instructions before installation, use, or maintenance of this appliance. Incorrect installation, use, or maintenance of this appliance can result in death, serious injury, or property damage. Keep these instructions with this appliance. This manual is subject to change. For the latest version, visit www.samsung.com.

Notices and notes

To make you aware of safety messages and highlighted information, we use the following notices and notes throughout this manual:



WARNING

Hazards or unsafe practices that may result in severe personal injury or death.



CAUTION

Hazards or unsafe practices that may result in minor personal injury or property damage.



IMPORTANT

Information of special interest



NOTE

Supplementary information that may be useful



The user and installer guides should be read carefully.



The user and installer guides should be read carefully.



WARNING

The installation and testing of this appliance must be performed by a qualified technician.

- The instructions in this manual are not intended as a substitute for proper training or adequate experience in the safe installation of the appliance.

Always install the air conditioner in compliance with current local, state, and federal safety standards.

General information

- The air conditioner should be used only for the applications for which it has been designed: the indoor unit is not suitable to be installed in areas used for laundry.

Installation of the product

- Our units must be installed in compliance with the spaces indicated in the installation manual to ensure either accessibility from both sides or ability to perform routine maintenance and repairs. The units' components must be accessible and that can be disassembled in conditions of complete safety either for people or things. For this reason, where it is not observed as indicated into the Installation Manual, the cost necessary to reach and repair the unit (in safety, as required by current regulations in force) with slings, trucks, scaffolding or any other means of elevation won't be considered in-warranty and will be charged to end user.

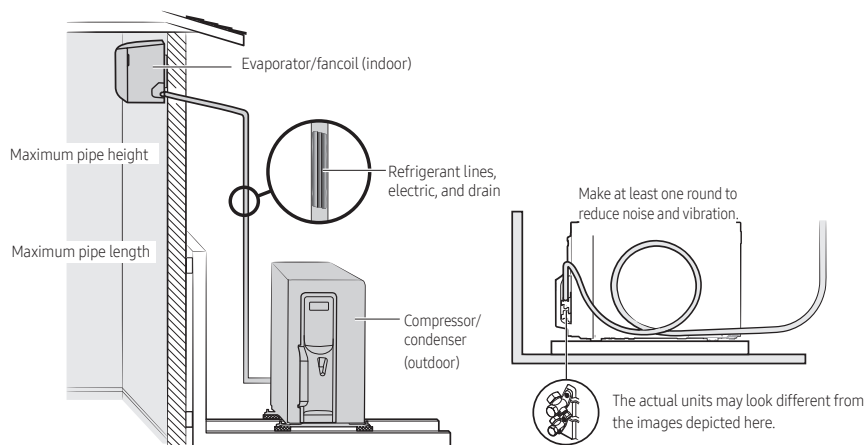
Power supply line, fuse, or circuit breaker

- Be sure not to perform power cable modification, extension wiring, and multiple wire connection.
 - It may cause electric shock or fire due to poor connection, poor insulation, or current limit override.
 - When extension wiring is required due to power line damage, refer to "Step 2-5 Optional: Extending the power cable" in the installation manual.

Preparation

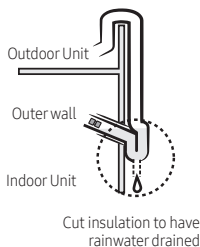
Step 1.1 Viewing the typical installation

A typical installation will be similar to the one shown below.



(Unit : m)

Model	Pipe length			Pipe height
	Minimum	Maximum	Standard for factory charge	Maximum
12TSHZ*	3	15	5	7
12TSFC*	3	15	5	8
18*** **24*****	3	30	5	15



Make a U-trap (A) on the pipe (which is connected to the indoor unit) at outer wall and cut the bottom part of the insulation (about 10 mm) to prevent rainwater from getting inside through the insulation.

Step 1.2 Choosing the installation location

WARNING

- Verify that a dedicated circuit breaker and a disconnect switch of the appropriate sizes for the air conditioner are preinstalled and available for use.
- Verify that the voltage and frequency of the power supply comply with the rated voltage as defined on the unit name plate.
- Verify that a suitable grounding connection is available.
- Do not install this appliance in an environment containing hazardous substances or close to equipment that releases open flames.
- Do not install this appliance near a heater or flammable material.

CAUTION

- The manufacturer shall not be responsible for damage occurring as a result of the wrong voltage being applied to this air conditioner.
- The indoor and outdoor units must be installed in compliance with minimum clearances to ensure that both units are accessible from both sides and can be maintained or repaired. Insufficient clearance may reduce product performance, generate excessive noise, and reduce the life of some unit components.

IMPORTANT

- Any changes or modifications to the installation described in this manual that are not expressly approved by the manufacturer could void the manufacturer's warranty.

To determine where to locate the indoor and outdoor units, you must survey the entire site and consider many variables. The goal is to select locations that comply with all safety precautions while also minimizing the total effort involved.

Indoor unit location requirements

WARNING

- Do not install the unit in a humid, oily, or dusty location or in a location exposed to direct sunlight, water, or rain.
- Make sure that the wall can support the unit weight.

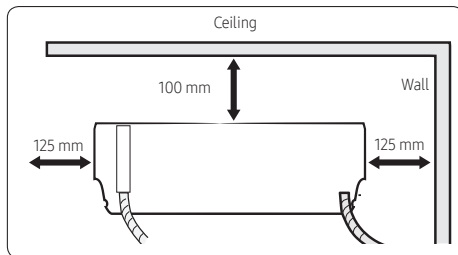
Examine the area that the customer wants to be air conditioned. Consider the following:

- What wall location will meet minimum clearances and provide optimal product performance?
- Will the wall provide adequate support for the unit weight (wall with stud construction or concrete)? If applicable, where are the studs?
- Where will you place the wall penetration for routing the piping bundle (consisting of power and communication cables, refrigerant pipes, and the drain hose) through the wall to the outdoor unit? Will the hole intersect any plumbing or wires in the wall?
- Is the location as close as possible to where the outdoor unit will be installed, to minimize the length of piping and cables?
- Will the condensate drain inside the room, through the wall penetration to the outdoor unit, or be connected to a condensate pump?

NOTE

- This manual covers a typical gravity-drain installation where the drain hose is routed to the outdoor unit through a hole in the wall.

Minimum clearances for the indoor unit



Preparation

Outdoor unit location requirements

Examine the area where the outdoor unit could be located. Consider the following:

- What location will meet minimum clearances and provide optimal product performance?
- Is there an existing level and hard foundation, such as a concrete pad, that will support the unit weight and produce minimal vibration? Installation on uneven ground may result in abnormal vibrations, noise, or problems with the unit.
- Does the unit need to be mounted on the wall?
- Where are the dedicated circuit breaker and disconnect switch located? How will you connect them to the unit?
- How will you route the piping bundle from the indoor unit? Is the location as close as possible to where the indoor unit will be installed, to minimize the length of piping and cables?
- Will the unit be sheltered from the wind? In a high-wind area, you may need to build a protective fence around the unit.
- Where will the condensate drain?



WARNING

- The drain location must allow condensate to drain properly and prevent ice from forming on the unit in winter. If a block of ice falls from the unit, it may result in death, serious injury, or property damage. Improper or inadequate draining may result in water overflowing and property damage.



CAUTION

- Do not connect the drain hose to existing waste pipes as odors may arise.

Installation on an exterior wall

If the outdoor unit must be installed on an exterior wall, you will need an L-bracket to support the unit. This bracket is not included with the unit.



WARNING

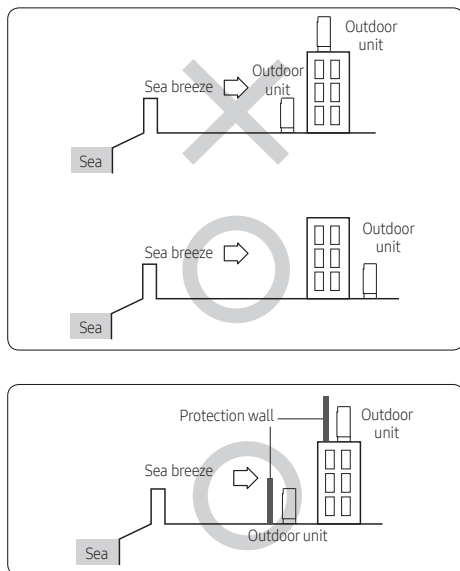
- The wall must be capable of supporting the weight of both the L-bracket and the outdoor unit. If the unit falls, it may result in crushing, electric shock, fire, or explosion that could cause death, severe personal injury, or property damage.

Installation Guide at the seashore

Make sure to follow below guides when installing at the seashore.

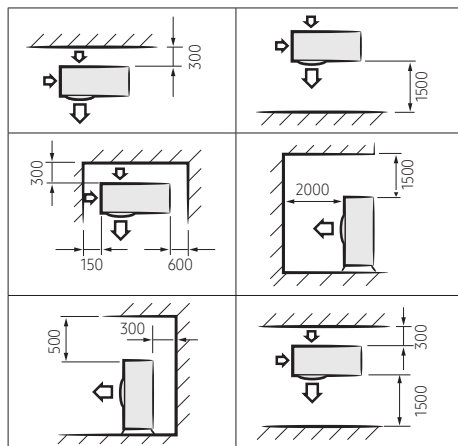
- 1 Do not install the product in a place where it is directly exposed to sea water and sea breeze.
 - Make sure to install the product behind a structure (such as building) that can block sea breeze.
 - Even when it is inevitable to install the product in seashore, make sure that product is not directly exposed to sea breeze by installing a protection wall.
- 2 Consider that the salinity particles clinging to the external panels should be sufficiently washed out.
- 3 Because the residual water at the bottom of the outdoor unit significantly promotes corrosion, make sure that the slope does not disturb drainage.
 - Keep the floor level so that rain does not accumulate.
 - Be careful not to block the drain hole due to foreign substance.
- 4 When product is installed in seashore, periodically clean it with water to remove attached salinity.
- 5 Make sure to install the product in a place that provides smooth water drainage. Especially, ensure that the base part has good drainage.
- 6 If the product is damaged during the installation or maintenance, make sure to repair it.
- 7 Check the condition of the product periodically.
 - Check the installation site every 3 months and perform anti-corrosion treatment such as R-Pro supplied by SAMSUNG (Code : MOK-220SA) or commercial water repellent grease and wax, etc., based on the product condition.
 - When the product is to be shut down for a long period of time, such as off-peak hours, take appropriate measures like covering the product.
- 8 If the product installed within 500m of seashore, special anti-corrosion treatment is required.

※ Please contact your local SAMSUNG representative for further details.



Examples for installing one outdoor unit:

Unit: mm



- Protection wall should be constructed with a solid material that can block the sea breeze and the height and width of the wall should be 1.5 times larger than the size of the outdoor unit. (You must secure more than 700 mm of space between the protection wall and the outdoor unit for air circulation.)

Minimum clearances for the outdoor unit

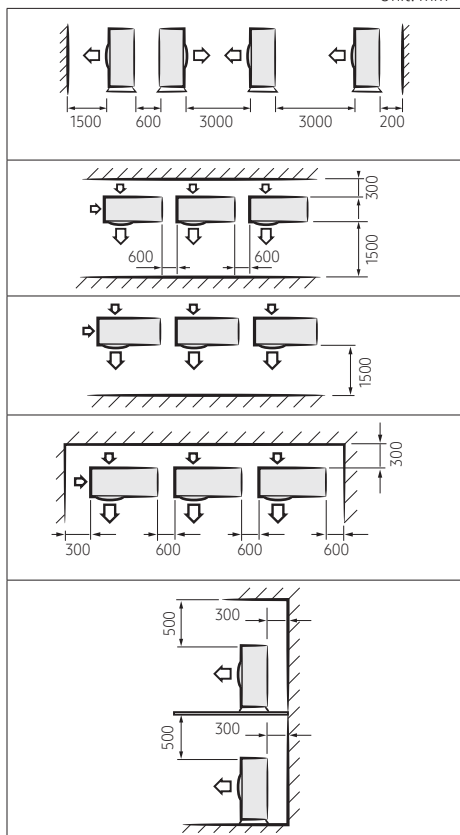
Legends:

Wall
Minimum clearance in mm
Air flow direction
Top view Side view

Preparation

Examples for installing multiple outdoor units:

Unit: mm



Step 1.3 Packing and Unpacking

Upon receipt, inspect the product to verify that it has not been damaged during transport. If the product appears damaged, do not install it and immediately report the damage to your local Samsung distributor.

Packing material must be disposed of in accordance with local regulations.

Indoor unit

Packing the unit as below procedure

- 1 Put the indoor unit onto the packing case.
- 2 Assemble cushion-both side of indoor unit. (Left and Right)
- 3 Seal the packing case.

Unpacking the unit as below procedure

- 1 Open the packing case.
- 2 Take out the both side of cushion(Left and Right).
- 3 Move the set from the packing case.

Outdoor unit

Packing the unit as below procedure

- 1 Put the outdoor unit onto the cushion-bottom.
- 2 Put the cushion-top onto the outdoor unit
- 3 Put the packing case from the top of set.
- 4 Seal the packing case.

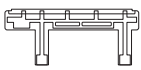
Unpacking the unit as below procedure

- 1 Take out the packing case from the set.
- 2 Take out the cushion-top.
- 3 Move the set from the cushion-bottom.

Step 1.4 Preparing materials and tools

Materials in the indoor unit package

Make sure that the indoor unit package contains the following materials:

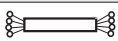
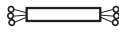
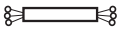
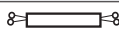
Mounting bracket (1) **12*****	Mounting bracket (1) **18/24*****
	
Remote control (1)	Remote control battery (2)
	
General information (1)	Quick guide (1)
	
Installation manual (1)	Holder remocon (1)
	
Extra M4 x12 tapping screw (2)	
	

Materials in the outdoor unit package

Make sure that the outdoor unit package contains the following materials:

Rubber foot (4)	Drain plug (1)
	

Optional accessories

Insulated assembly pipe, Ø 6.35 mm (1)	Insulated assembly pipe, Ø 9.52 mm (1) **12*****
	
Insulated assembly pipe, Ø 12.70 mm (1) **18*****	Insulated assembly pipe, Ø 15.88 mm (1) **24*****
	
Pipe clamp B (3)	Pipe clamp A (3)
	
Drain Hose, 2 m long (1)	Foam Insulation (1)
	
Vinyl tape (2)	PE T3 foam tube insulation (1)
	
Putty 100 g (1)	M4 x 25 tapped screw (6)
	
Cement nail (6)	4-wire Assembly Cable (1) **12TSHZ***
	
3-wire Power Cable (1) **12TSFC*** **18/24*****	3-wire Assembly Cable (1) **12TSFC*** **18/24*****
	
2-wire Assembly Cable (1) **12TSFC*** **18/24*****	
	

Preparation

NOTE

- A flare nut is attached to the end of each refrigerant pipe coming from the evaporator. Use these flare nuts when connecting the pipes.

Materials supplied by the installer

Make sure you have all other materials required for the selected installation method and location.

IMPORTANT

- No mounting hardware, tubing, cables, and other materials listed below are included with the appliance.

The required materials will vary, but may include the following:

- 1.8 m electrical whip for connecting the power from the installed disconnect switch to the outdoor unit
- UV-resistant vinyl line set tape for the exposed line set
- Lines-set cover and fittings, if used
- Miscellaneous pipe hangers
- Miscellaneous screws and anchors for hanging pipe hangers, the line-set cover, the indoor unit mounting bracket, and so on.
- Electrical ring connectors for connecting all power and communication wiring
- Electrical tape
- Refrigerant R-410A if additional refrigerant is required due to line-set length
- Closed cell foam tape insulation (roll)
- Outdoor unit risers or L-brackets for wall installation
- Silicone caulking for sealing the wall penetration
- Rags

Tools

Make sure you have the required tools available.

General tools

- Vacuum pump
(Backward flowing prevention)
- Manifold gauge
- Stud finder
- Torque wrench
- Pipe cutter
- Reamer
- Pipe bender
- Spirit level
- Screwdriver
- Spanner
- Drill
- L-wrench
- Measuring tape

Tools for test operation

- Thermometer
- Resistance meter
- Electroscopes

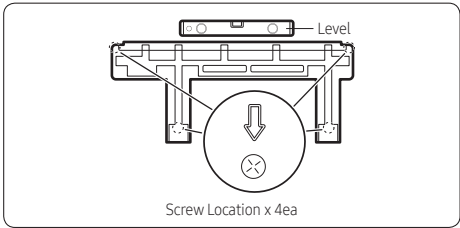
Indoor Unit Installation

Step 2.1 Attaching the mounting bracket to the wall

- 1 Hold the mounting bracket against the wall at the selected installation position (Step 1.2 on page 5), making sure that the screw holes align with the center of the studs in the wall. If the screw locations do not align with the studs, use wall anchors.

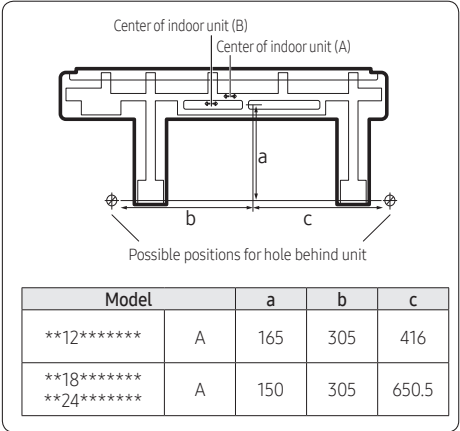
CAUTION

- The recommended best practice is to attach the mounting bracket directly to the studs in the wall. If you did not find a suitable location with studs (in Step 1.2 on page 5), or if the wall is concrete, you must use wall anchors of a suitable type and weight capacity, and install them according to the manufacturer's instructions. Failure to do so may cause the material surrounding the joints to crumble over time and the screws to be loosened and stripped. This may result in the unit falling from the wall, which could cause physical injury or equipment damage.
- 2 Using a level, make sure that the mounting bracket is level, then mark the location of the screw holes on the wall.
 - 3 If using wall anchors, install them at the screw hole positions, following the manufacturer's instructions.
 - 4 Using six field-supplied mounting screws and anchors (if applicable), attach the bracket to the wall.



Step 2.2 Drilling the wall penetration

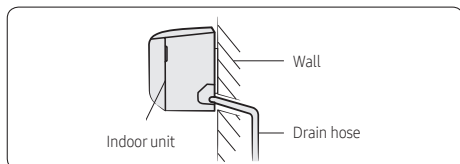
- 1 Determine the position of the hole through which the piping bundle (consisting of power and communication cables, refrigerant pipes, and the drain hose) will pass. Consider the following:
 - The hole inner diameter must be 65 mm.
 - The recommended hole location is behind the unit so that the hole and the piping bundle will not be visible in the room. The minimum distances between the hole and the mounting bracket are:



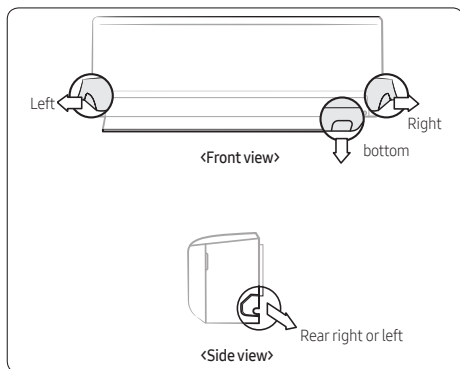
- If the hole cannot be positioned behind the unit, find a position as close to the unit as possible. The piping bundle that exits the unit and extends to the hole will need to be attached to the wall and will be visible inside the room.
- In relation to the bracket shown above, the unit is shipped with the drain hose connection on the right, the drain hose exits the unit and extends to the right, and the refrigerant pipes are bent to exit on the left. Thus, positioning the hole to the left requires the least effort. If you position the hole to the right or below the unit, you will need to move the drain hose connection to the left and bend the pipes so that the hose and pipes exit to the right or bottom. See the figure in step 3 on page 12.

Indoor Unit Installation

- 2 Use a standard 65 mm hole saw to drill one hole at the selected location, at a 15° downward angle so that the drain hose will drain properly.



- 3 Based on the hole location, determine where the piping bundle (drain hose, refrigerant pipes, and cables) will exit the unit.



NOTE

- The left, right, or bottom exit will only be used if the hole is not positioned behind the unit.

Step 2.3 Connecting the refrigerant pipes

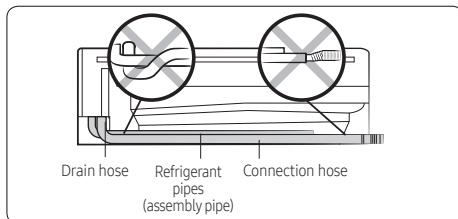
Connect indoor and outdoor units with field-supplied copper pipes by means of flare connections. Use insulated seamless refrigeration grade pipe only, (Cu DHP type according to ISO1337), degreased and deoxidized, suitable for operating pressures of at least 4200 kPa and for burst pressure of at least 20700 kPa. Under no circumstances must sanitary type copper pipe be used.

IMPORTANT

- When installing the unit, always connect the refrigerant pipes first, followed by the electrical cables. For disassembly, always disassemble the electric cables before the refrigerant pipes.

Two short refrigerant pipes are already attached to the air conditioner:

- The smaller-diameter pipe is for the high-pressure, two-phase refrigerant.
- The larger-diameter pipe is for the low-pressure refrigerant vapor.



In Step 2.2, step 3 you determined the exit position for the piping bundle. The unit has three knockouts available for the left, right, and bottom exits. When the bundle exits directly from the rear, none of the knockouts are used.

- 1 If the pipes will exit directly from the rear, skip to step 3. Otherwise, cut out the appropriate knockout piece (left, right, or bottom).

- 2 Use a razor knife to clean the cut edges (flashing).

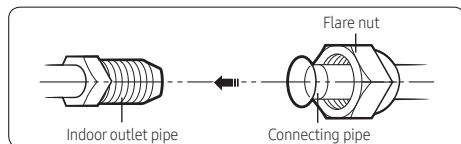
- 3 The left exit is the only position that does not require bending the pipes. For other positions, bend the pipes so that they will exit in the selected exit position.

- The bending radius should be greater than 100 mm.
- Bend the smaller pipe gradually to prevent kinking. The larger pipe has a preinstalled spring bender to prevent kinking.
- Make sure that the pipes do not protrude from the back of the unit in a way that will make it difficult to attach the unit to the mounting bracket.
- For right and bottom exits, pull the pipes out through the selected knockout opening. For left exits, the piping connections will be made in the service space behind the indoor unit (under the cover panel).

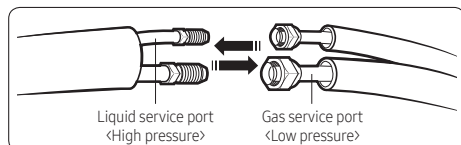
NOTE

- If you are using the right rear exit, the pipes should be long enough to extend through the wall without needing to connect the line set first. It may be easier to connect the line set outside of the building, after you have bundled the pipes and cables and passed the bundle through the wall. In this case, do not connect the line set now. Instead, complete Step 2.4 through Step 2.7, then go outside and connect the line set as described below.

- 4 Slowly remove the protective caps on the refrigerant pipe connections to relieve the nitrogen holding charge.
- 5 Connect the line set to each pipe.



- 6 Hand-tighten the flare nuts to make sure that they do not become stripped.



- 7 Torque the flare connections to the following values:

(Unit: mm)

Outer diameter	Torque (N·m)
ø 6.35	14–18
ø 9.52	34–42
ø 12.70	49–61
ø 15.88	68–82



CAUTION

- Tighten the flare nuts only to the specified torque. If a flare nut is overtightened, the flare face may crack, causing refrigerant leakage.
- 8 Do not box in or cover the pipe connections. Make sure that the connections are accessible for testing later in the installation process and for future servicing.
 - 9 Tape over the end of the pipes so that debris will not enter the piping when it is passed through the wall. The pipes will be insulated later in the installation process.

Step 2.4 Connecting the power and communication cables



WARNING

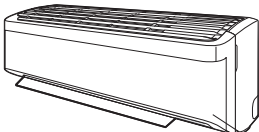
- Do not modify the power cable in any way. Doing so may cause electric shock or fire due to poor connection, poor insulation, or current limit override. Make sure to comply with the technical standards of electrical installations and the wiring regulations in the local area.
- This appliance must be properly grounded. Do not ground the appliance to a gas pipe, plastic water pipe, or telephone line. Failure to comply may result in electric shock, fire, and explosion.

- 1 Connect each wire to its corresponding terminal number.

Model	**12TSHZ***
Power cable (Indoor unit)	3G X 1.0 mm ² , H05VV-F
Indoor-to-outdoor power cable	4G X 1.0 mm ² , H07RN-F
Type GL	16A

Model	**12TSFC***	**18***** **24*****
Power cable (Outdoor unit)	3G X 2.5 mm ² , H07RN-F	3G X 2.5 mm ² , H07RN-F
Outdoor-to-indoor power cable	3G X 1.0 mm ² , H07RN-F	3G X 1.0 mm ² , H07RN-F
Communication cable	2 X 0.75 mm ² , H05RN-F	2 X 0.75 mm ² , H05RN-F
Type GL	16A	20A

Indoor Unit Installation



Control box

Before connecting				
	Correct	Upside down	Damaged	Non-circular
After connecting				
	Correct (Front view)	Correct (Side view)	Upside down	Non-fitted

<Circular terminal>

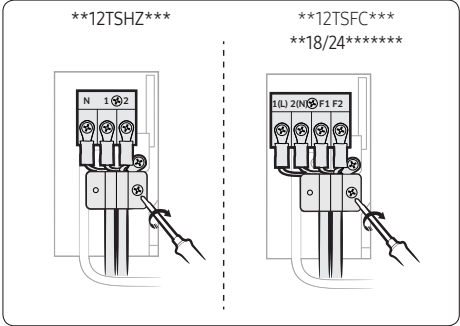
CAUTION

- Connect the wires firmly so that wires cannot be pulled out. Loose wires can cause the connection to overheat.
- Each circular terminal must match the size of its corresponding screw in the terminal block.

CAUTION

- For the terminal block wiring, use a wire with a ring terminal socket only. Regular wires without a ring terminal socket may become a hazard as the connections may loosen during operation.

2 Tighten the terminal block screw.



3 In Step 2.2, step 3 you determined the exit position for the piping bundle. If using the left, right, or bottom exits, pass the cables through the selected knockout.

NOTE

- Power supply cords of parts of appliances for outdoor use shall not be lighter than polychloroprene sheathed flexible cord. (Code designation IEC: 60245 IEC66/ CENELEC: H07RN-F, IEC: 60245 IEC57 CENELEC: H05RN-F, IEC: 60227 IEC53: H05VV-F)
- Power & Communication cable shall not exceed 30 m.

Step 2.5 Optional: Extending the power cable

- 1 Prepare the following tools.

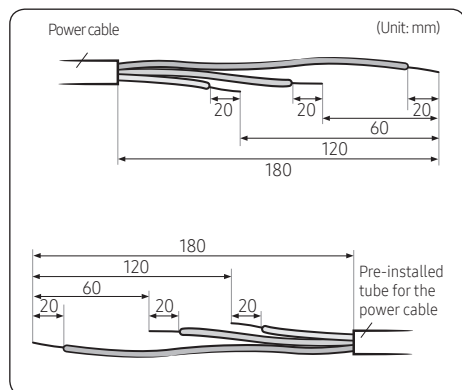
Tools	Spec	Shape
Crimping pliers	MH-14	
Connection sleeve (mm)	20xØ6.5 (HxOD)	
Insulation tape	Width 19 mm	
Contraction tube (mm)	70xØ8.0 (LxOD)	

- 2 As shown in the figure, peel off the shields from the rubber and wire of the power cable.
- Peel off 20 mm of cable shields from the pre-installed tube.



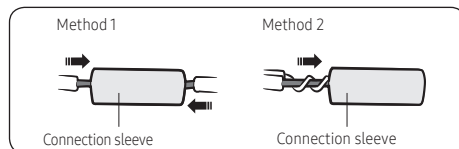
CAUTION

- For information about the power cable specifications for indoor and outdoor units, refer to the installation manual.
- After peeling off cable wires from the pre-installed tube, insert a contraction tube.
- If cable wires are connected without using connecting sleeves, their contact area becomes reduced, or corrosion develops on the outer surfaces of the wires (copper wires) over a long time. This may cause an increase of resistance (reduction of passing current) and consequently may result in a fire.



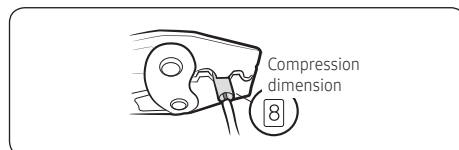
- 3 Insert both sides of core wire of the power cable into the connection sleeve.

- Method 1:** Push the core wire into the sleeve from both sides.
- Method 2:** Twist the wire cores together and push it into the sleeve.

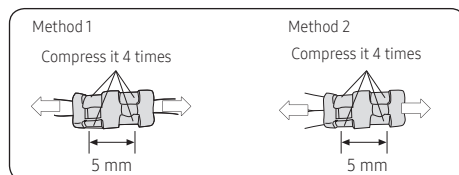


- 4 Using a crimping tool, compress the two points and flip it over and compress another two points in the same location.

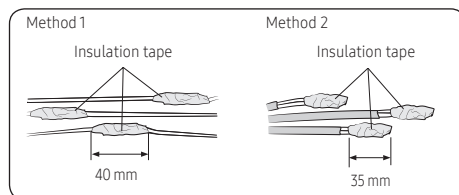
- The compression dimension should be 8.0.



- After compressing it, pull both sides of the wire to make sure it is firmly pressed.

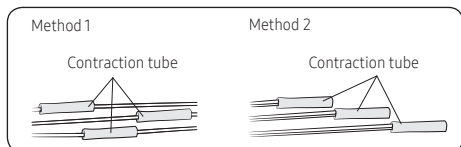


- 5 Wrap it with the insulation tape twice or more and position your contraction tube in the middle of the insulation tape.

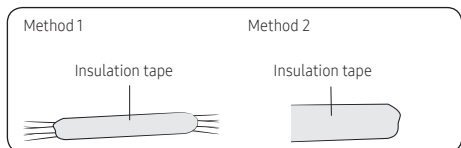


Indoor Unit Installation

- 6 Apply heat to the contraction tube to contract it.



- 7 After tube contraction work is completed, wrap it with the insulation tape to finish.
Three or more layers of insulation are required.

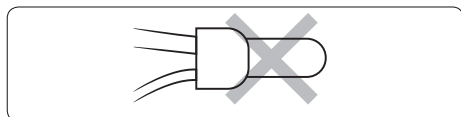


CAUTION

- Make sure that the connection parts are not exposed to outside.
- Be sure to use insulation tape and a contraction tube made of approved reinforced insulating materials that have the same level of withstand voltage with the power cable. (Comply with the local regulations on extensions.)

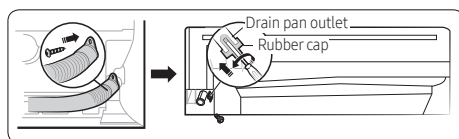
WARNING

- In case of extending the electric wire, please DO NOT use a round-shaped Pressing socket.
 - Incomplete wire connections can cause electric shock or a fire.



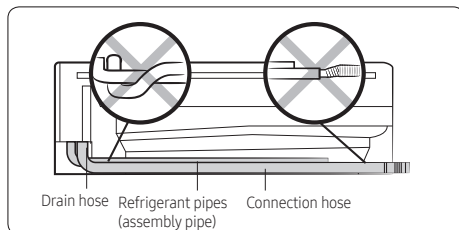
Step 2.6 Connecting the drain hose

- 1 In Step 2.2, step 3 you determined the exit position for the piping bundle. If using the right, bottom, or right rear exit, change the drain hose connection from the right to the left so that the drain hose will lie along the inside of the unit and exit to the right.



CAUTION

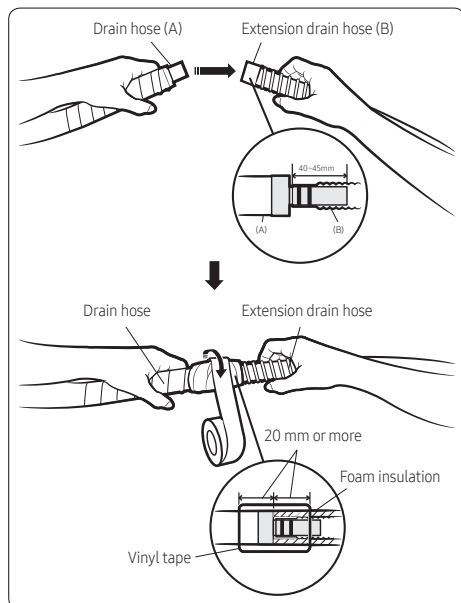
- Be careful not to puncture the plug with the screwdriver when installing it.
- 2 If using the left, right, or bottom exit, pass the drain hose through the selected knockout.



- 3 Connect a 15.88 mm ID extension drain hose to the main drain hose.

⚠ CAUTION

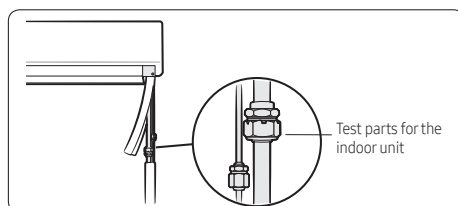
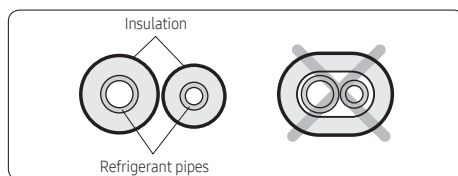
- If the diameter of the connection hose is smaller than the product's drain hose, leakage may occur.



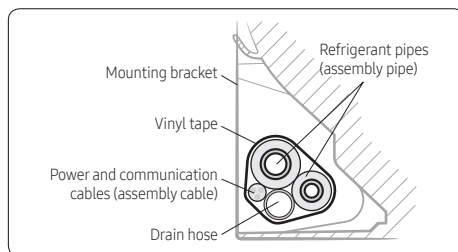
- 4 Do not box in or cover the drain hose connection. It must be accessible for testing later in the installation process and for future servicing.
- 5 If the drain hose is routed inside the room, insulate the hose so that dripping condensation does not damage the furniture or floors.

Step 2.7 Taping the pipes, cables, and drain hose

- 1 Wrap foam insulation around the refrigerant pipes, up to the connection points. The connections must remain accessible for testing later in the installation process. Either leave slits in the insulation or do not cover the connections.



- 2 Make a piping bundle by using vinyl tape to wrap together the refrigerant pipes, power cable, communication cable, and drain hose, up to the connection points. Connection points must remain accessible for testing later in the installation process.



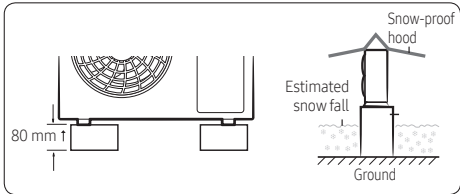
Outdoor Unit Installation

Step 3.1 Mounting the outdoor unit

To promote proper condensate draining, the recommended installation of the outdoor unit is elevated above the ground on a mounting bracket attached to a concrete pad.

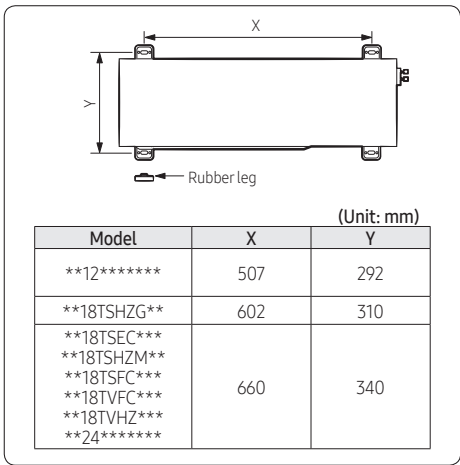
In areas where snowfall occurs, the unit must be mounted above the snow line to allow for proper heating. Snow cannot be allowed to collect on top of the unit. For promoting natural drainage in a heavy snow fall area:

- Make space more 80 mm between the bottom of the outdoor unit and the ground for installation. (Ensure that the drained water runs off correctly and safely.)
- Allow enough separation distance between the product and the ground.



On the ground

- 1 Place the outdoor unit in the selected installation location (Step 1.1 on page 4), ensuring proper clearances and with the arrow on top of the unit pointing away from the wall.
- 2 Clip the rubber feet to the tabs to minimize sound and vibration to the structure.

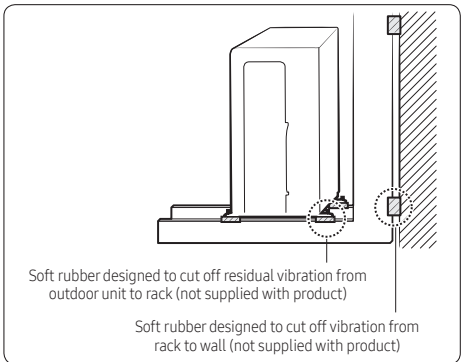


- 3 Level the unit, then use anchor bolts to secure it at the four mounting points.
- 4 For installations in locations that require seismic or hurricane tie downs, comply with local codes.
- 5 If the selected location is exposed to strong winds, install a protective fence around the unit so that the fan can operate correctly.

On a wall

WARNING

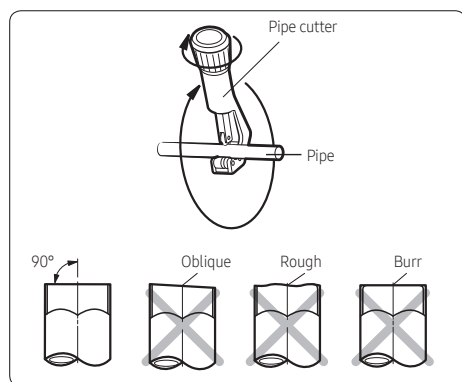
- The unit must be properly secured to the wall. If the unit falls, it may result in crushing, electric shock, fire, or explosion that could cause death, severe personal injury, or property damage.
- 1 At the selected installation location (Step 1.1 on page 4), attach the L-bracket to the wall as follows:
 - Install the bracket as close to the wall as possible.
 - Insert rubber isolators between the bracket and the wall to minimize sound and vibration to the structure. Do not fully compress the isolators.



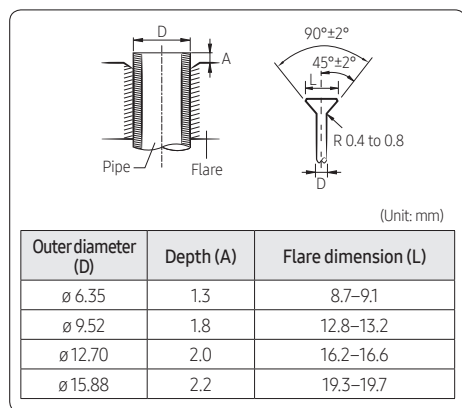
- Make sure that the bracket is level.
 - Use suitable bolts/washers and lock washers.
- 2 Place the outdoor unit on the bracket, ensuring proper clearances and with the arrow on top of the unit pointing away from the wall.
 - 3 Clip the rubber feet to the tabs to minimize sound and vibration to the structure.
 - 4 Level the unit, then use anchor bolts to secure it at the four mounting points.
 - 5 For installations in locations that require seismic or hurricane tie downs, comply with local codes.

Step 3.2 Connecting the cables and the pipes

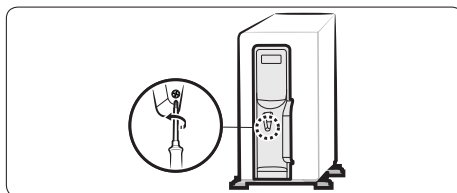
- 1 Route the piping bundle to the outdoor unit.
- 2 Use piping clamps to fasten the piping bundle to the foundation or wall.
- 3 Cut the refrigerant pipes to the length needed to reach the pipe connections (located behind the cover panel; see the figure in step 7).



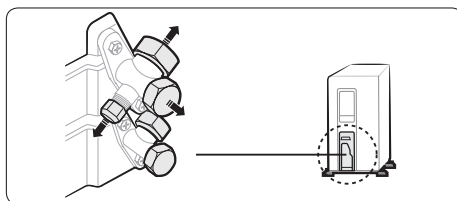
- 4 Remove any burrs, positioning the pipe face down to make sure that the burrs do not get into the pipe.
- 5 Assemble the flare connections on the cut pipe ends.



- 6 Remove the cover panel on the unit.

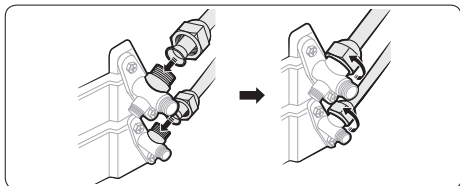


- 7 Remove the service valve caps.

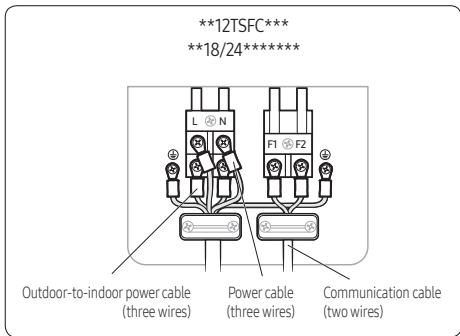
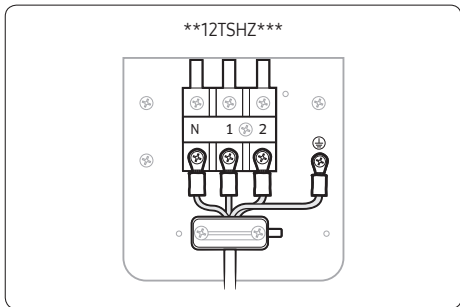


Outdoor Unit Installation

- 8 Connect the pipes to the service valve with the flare nuts. Hand-tighten the nuts to prevent stripping.



- 9 Torque the flare connections to the values in Step 2.3, step 7 on page 13.
- 10 Connect the power cables and secure with a cable clamp.

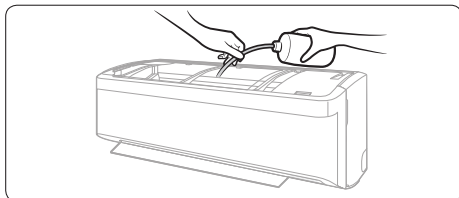


- 11 Connect the outdoor unit power supply cable to the preinstalled disconnect switch.
- 12 Leave the cover panel off for testing later in the installation process.

Installation Inspection and Testing

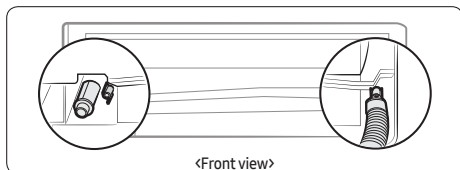
Step 4.1 Performing a drain leak test

- 1 Pour water into the drain pan.



CAUTION

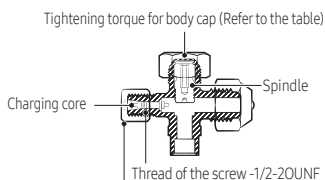
- Make sure that the water does not overflow onto the electrical connection.
- 2 Check for leaks at the drain connection under the cover panel.



- 3 Make sure that the hose is draining properly at the outdoor unit.

Step 4.2 Performing the gas leak tests

- 1 Before inspecting the leakage, use a torque wrench to close the cap for the stop valve. (Comply with a tightening torque for each size of the diameter, and tighten the cap firmly to prevent any leakage.)

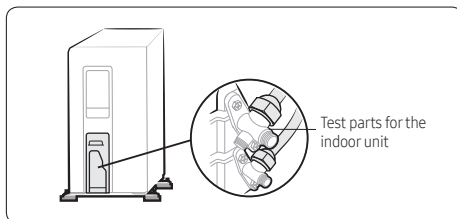
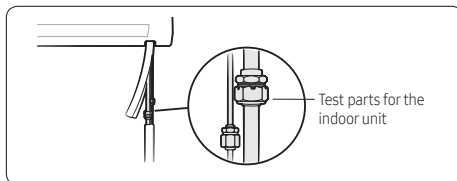


Tightening torque for charging port cap (Refer to the table)

Outer diameter (mm)	Tightening torque	
	Body cap (N•m)	Charging port cap (N•m)
ø 6.35	20 to 25	10 to 12
ø 9.52	20 to 25	
ø 12.70	25 to 30	
ø 15.88	30 to 35	
Over ø 19.05	35 to 40	

(1 N•m = 10 kgf•cm)

- 2 Insert inert gas into the pipes connected to indoor and outdoor units.
- 3 Test leakage on the connection parts of the indoor and outdoor units with soap lather or liquid.



Installation Inspection and Testing

Step 4.3 Evacuating the system

CAUTION

- Because the system does not have filter driers, you must perform this triple evacuation procedure to remove all noncondensables and moisture from the system before charging. Failure to do so will result in reduced performance and shorter equipment life.

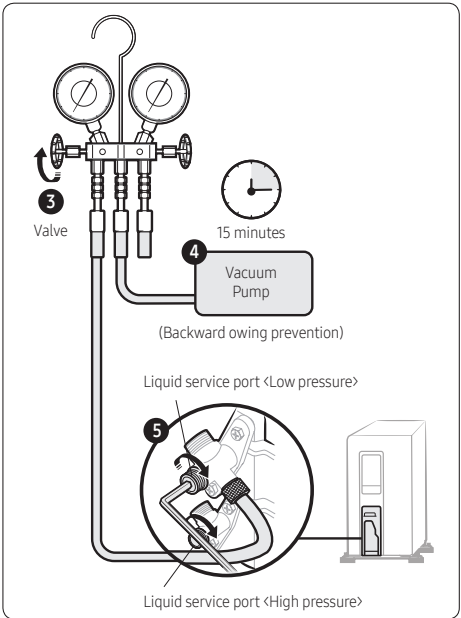
The time required to perform each evacuation will depend on the capacity (CFM) of the vacuum pump used.

- 1 Install a micron vacuum gauge to the larger liquid/vapor line's service port on the branch of a tee.
- 2 Install the red high-side hose of an R-410A gauge manifold set to the smaller liquid/vapor line's service port on the run of the tee.
- 3 Attach a vacuum pump to the common hose of the manifold set.
- 4 To ensure optimal performance, verify that the vacuum pump's oil has been changed recently.
- 5 With the service port closed and the manifold gauge open, start the vacuum pump and make sure that the vacuum level drops below 4000 microns (as read on the micron gauge). If it is difficult to achieve a proper vacuum, a leak in the hoses is likely. Repair the leak(s) and/or check performance of vacuum pump, then repeat this step.
- 6 Open the service port to connect the system to the manifold.
- 7 Evacuate until 4000 microns is achieved, for at least 10 minutes.
- 8 Close the gauge manifold valve, shut off the vacuum pump, and remove the common hose.
- 9 Connect the hose to the nitrogen pressure regulator and bleed the hose by opening the end of the common hose closest to the manifold.
- 10 Open the high-pressure manifold valve and slowly bring the system pressure to atmosphere (50 kPa).
- 11 Close the manifold and nitrogen cylinder and remove the common hose.

- 12 Reconnect the common hose to the vacuum pump. Repeat steps 6 through 12, alternating between breaking the vacuum with dry nitrogen and evacuating, until system evacuation has occurred three times, to the following vacuum levels:

Evacuation	Microns
First	4000
Second	2000
Third	500

- 13 After evacuating to at least 500 microns for the third time, close the gauge manifold valve and wait 10 minutes, making sure that the vacuum level in the system does not decrease. If it does, a small leak is likely. Repair the leak and repeat the evacuation process.



Step 4.4 Adding refrigerant (if needed)

The outdoor unit is charged with sufficient R-410A refrigerant to support up to a 5 m line set. For lengths greater than 5 m, you must add 15g of refrigerant per meter of additional length, after the lines are evacuated.

- 1 Calculate the additional refrigerant required:
Additional grams of R-410A = (Total line set meter – 5) × 15
- 2 Connect the common hose of the manifold gauge set to the inverted R-410A refrigerant cylinder.
- 3 Place the refrigerant cylinder on a scale set to measure grams.
- 4 Open the valve on the tank.
- 5 At the manifold connection, bleed the refrigerant to remove any air that may be present in the common hose.
- 6 Open the gauge manifold and charge the system with the amount of refrigerant calculated in step 1.
- 7 Close the gauge manifold valve, close the valve on the refrigerant tank, and remove the common hose.

Step 4.5 Preparing the system for commissioning

- 1 Wrap the remaining refrigerant pipe lengths and connection points with foam insulation.
- 2 Wrap the unwrapped portions of the piping bundle with vinyl tape.
- 3 With the manifold gauge set still installed, open the isolation valves on the outdoor unit to connect the outdoor unit to the line set and indoor unit.
- 4 Remove the manifold set and vacuum gauge.

Step 4.6 Commissioning the unit

The unit is commissioned using the Smart Install feature. Smart Install can be started only with the remote control. While Smart Install is running, you cannot operate the remote control.

- 1 Make sure that the air conditioner is in standby status (powered up with the controller in off mode).
- 2 Install batteries in the remote control.
- 3 Hold down the  (Power),  (Mode), and  (SET) buttons on the remote control simultaneously for 4 seconds.
- 4 Wait until Smart Install succeeds or fails (approximately 7 to 13 minutes).
 - While Smart Install is running:

Type	88Display
Indoor unit indicator	
	The progress is displayed as a number between 0 and 99 on the indoor unit display.

- When Smart Install succeeds: Smart Install ends with a ringing sound, and the air conditioner returns to standby status.
- When Smart Install fails: An error message is displayed on the indoor unit display, and Smart Install ends. To correct the problem, see the error table on page 24.

Installation Inspection and Testing

Error indicator	Error	Measures for the installer to take
 88 Display		
 C101	Communication error between indoor and outdoor units	Check the cables between the indoor and outdoor units. See if the power cable or communication cable is crossed.
 C121	Error on indoor temperature sensor	Make sure that the indoor temperature sensor is properly connected.
 C122  C123	Error on indoor heat exchanger	Make sure that the evaporator temperature sensor is properly connected.
 C154	Error on indoor fan motor	- Make sure that the evaporator motor is properly connected to the board. - Check for a foreign substance inside the unit that may be preventing the blower wheel from turning.
 88  C162  C163	EEPROM/Option error	Reset the option codes.
 C422	Refrigerant flow blocking error	- Make sure that the service valves are completely open. - Check for any blockage in the refrigerant pipe that connects the indoor and outdoor units. - Check for refrigerant leaks.
 C554	Lack of refrigerant	- Make sure a sufficient amount of refrigerant has been added for a pipe that is longer than 7.5 m. - Check for refrigerant leaks between the valve and pipe connection.

Step 4.7 Performing final checks and trial operation



WARNING

Stop the unit, disconnect the power, and contact Samsung technical support if any of the following occurs:

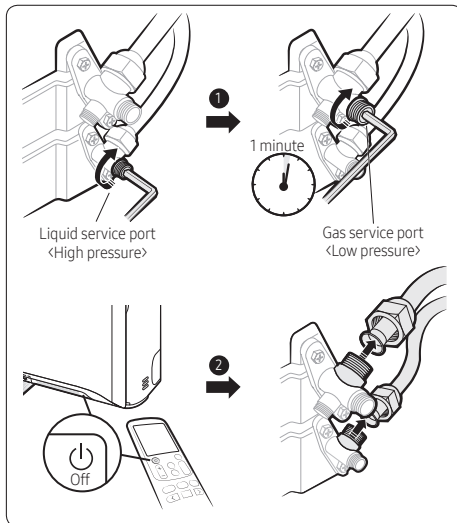
- The unit produces a burning smell or smoke.
- The power cable is hot or damaged.
- The unit is very noisy.
- Any foreign substance, such as water, has entered the appliance.
- The appliance becomes flooded.

- 1 Check the following:
 - Strength of the installation site
 - Tightness of pipe connection to detect gas leak
 - Electric wiring connection
 - Heat-resistant insulation of the pipe
 - Drainage
 - Grounding conductor connection
 - Correct operation (Take the following steps.)
- 2 Press the  (Power) button on the remote control to check the following:
 - The indicator on the indoor unit lights up.
 - The airflow blade opens and the fan gears up for operation.

- 3 Press the  (Mode) button to select Cool or Heat mode. Then take the following sub-steps:
 - In Cool mode, use the Temperature button to set the set temperature to 16 °C.
 - In Heat mode, use the Temperature button to set the set temperature to 30 °C.
 - Check whether, approximately 3 to 5 minutes later, the outdoor unit starts, and a cool or warm air blows out.
 - After 12 minutes of stationary condition, check the indoor unit air treatment.
- 4 Press the  (Air swing) button to check whether the airflow blades work properly.
- 5 Press the  (Power) button to stop the trial operation.

CAUTION

- Compressor damage may occur if the compressor is run at a negative suction pressure.



Pumping down for removing the product

Pump-down is an operation intended to collect all the system refrigerant in the outdoor unit. This operation must be carried out before disconnecting the refrigerant tubing in order to avoid refrigerant loss to the atmosphere.

WARNING

- After installing the product, be sure to perform leak tests on the piping connections. After pumping down refrigerant to inspect or relocate the outdoor unit, be sure to stop the compressor and then remove the connected pipes.
 - Do not operate the compressor while a valve is open due to refrigerant leakage from a pipe or an unconnected or incorrectly connected pipe. Failure to do so may cause air to flow into the compressor and a too high pressure can develop inside the refrigerant circuit, leading to an explosion or product malfunction.
- 1 Hold down the  (Power) button on the indoor unit for 5 seconds. Beep sounds immediately to indicate that the product is ready for pump down procedure.
 - 2 Let the compressor run for more than 5 minutes.
 - 3 Release the valve caps on High and Low pressure side.
 - 4 Use L-wrench to close the valve on the high pressure side.
 - 5 After approximately 1 minute, close the valve on the low pressure side.
 - 6 Stop operation of the air conditioner by pressing the  (Power) button on the indoor unit or remote control.
 - 7 Disconnect the pipes.

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